

Research Suggestions for AGN SURD Paper and Databases

Compiled from the two previous answers, preserving the full content, tables, equations, and recommendations.

Research-paper suggestions based on the AGN SURD notebook

1. Study of the problem and code: what else is needed for a research paper

I studied the uploaded notebook. My assessment is: it is a promising pilot analysis, but not yet a publishable research paper. The main idea is good: use SURD information decomposition to go beyond standard pairwise reverberation mapping and ask whether AGN emission-line variability is driven by unique, redundant, or synergistic information from continuum and line components. SURD is a recent causal-information method that decomposes information about future states into unique, redundant and synergistic contributions. Source note: Nature Communications, "Decomposing causality into its synergistic, unique, and redundant components".

1.1. What the current notebook already does well

The notebook already contains the skeleton of a paper:

It defines a clear motivation: standard reverberation mapping usually estimates lags by cross-correlation, while SURD can ask a richer question: whether a target light curve is predicted uniquely by one driver, redundantly by several drivers, or synergistically by their combination.

It applies the idea to NGC 5548, using continuum and H β data. This is a good target because NGC 5548 has a long reverberation-mapping history and velocity-resolved broad-line-region studies. Recent work continues to show that NGC 5548 is dynamically complex, with evolving BLR structure and velocity-resolved H β information.

The code also goes beyond a single calculation. It includes:

- synthetic AGN-like tests,
- H β velocity-bin construction: blue wing, core, red wing,
- lag scans,
- histogram-bin robustness tests,
- cadence robustness tests,
- circular-shift surrogate tests,
- extended lag scans up to 120 days,
- summary tables of maximum synergy and minimum information leak.

This is a good start.

1.2. The most important weakness: the physical interpretation is not yet safe

The notebook finds, for example, peak synergy around:

Target	Peak synergy lag, first scan	Extended scan result
Core H β	52 days	111 days
Red wing H β	59 days	93 days
Blue wing H β	43 days	no longer peak beyond 60 days

This is interesting, but it is also dangerous. The classical H β reverberation lag of NGC 5548 is often much shorter, typically of order tens of days, and recent 2023 monitoring found an H β rest-frame lag around 15.6 days.

So the long SURD synergy peaks should not yet be interpreted directly as reverberation lags. They may instead reflect:

- long-timescale autocorrelation of the continuum,
- seasonal sampling structure,
- interpolation artifacts,
- hidden common drivers,
- BLR breathing or secular variability,
- nonlinear coupling between velocity components,
- or insufficient conditioning on the target's own past.

For a research paper, this distinction is essential:

The SURD synergy peak is not automatically the same object as the ICCF/JAVELIN reverberation lag.

This should become one of the central methodological points of the paper.

1.3. A serious code issue: the early notebook uses a duplicated continuum signal

In the first real-data SURD analysis, the notebook uses:

$$\begin{aligned} S1 &= \text{continuum}, \\ S2 &= \text{continuum copy}, \\ S3 &= \text{H}\beta. \end{aligned}$$

This is useful only as a sanity check. It should not be presented as a real multi-driver AGN result, because the second predictor is not an independent physical channel.

For the paper, you need real physically distinct predictors, for example:

$$\begin{aligned} S1 &= 5100 \text{ \AA continuum}, \\ S2 &= \text{UV continuum, X-ray continuum, or another optical band}, \\ T &= \text{H}\beta \text{ total flux or one velocity bin}. \end{aligned}$$

If only one continuum band is available, then a better paper design is:

$$\begin{aligned} S1 &= \text{continuum}, \\ S2 &= \text{one H}\beta \text{ velocity component}, \\ T &= \text{another H}\beta \text{ velocity component}. \end{aligned}$$

This is what the later part of the notebook begins to do, and that direction is much more publishable.

1.4. The MJD/time-handling must be cleaned carefully

The notebook has evidence of an important time-axis correction. At one point, the velocity-resolved spectra initially appear with an MJD range around 97512-99255, then later the corrected range is 47512-49255. This is not a small bookkeeping detail. In reverberation mapping, a wrong MJD offset destroys the physical lag interpretation.

For the paper, you need a clean data section with a table like:

Data product	Number of points	MJD range	Median cadence	Source
5100 Å continuum	...	...	...	AGN Watch/STORM
integrated H β	...	...	...	AGN Watch/STORM
H β blue wing	...	...	...	processed from spectra
H β core	...	...	...	processed from spectra

Data product	Number of points	MJD range	Median cadence	Source
H β red wing	...	...	...	processed from spectra

You also need to state exactly how the filename-to-MJD conversion is done and verify it against the original campaign documentation.

1.5. The interpolation method needs improvement

The current code interpolates irregular light curves onto a uniform grid. That is acceptable for a first test, but for a paper it is not enough.

Problems:

1. AGN light curves are irregularly sampled.
2. Interpolation can create artificial temporal structure.
3. Forward/backward filling can introduce false correlations.
4. SURD results may depend on the imposed cadence.

You already tested cadences of 1, 2 and 5 days, which is good. But the paper should add at least two more checks:

First, compare with a method that avoids aggressive interpolation, for example using only overlapping observed epochs within a lag tolerance.

Second, use Monte Carlo interpolation uncertainty: perturb fluxes using measurement errors, resample the light curves many times, recompute SURD, and report confidence bands.

1.6. The null tests need to be stronger

The current circular-shift surrogate test is useful, but not sufficient. Circular shifts destroy phase alignment but preserve some autocorrelation. For AGN light curves, the null model must preserve red-noise behavior and sampling structure.

For a publishable result, I would add:

Null test	Purpose
Circular-shift surrogate	Tests whether alignment matters
Block bootstrap	Preserves local time correlation
Phase-randomized / IAAFT surrogate	Preserves power spectrum approximately
Flux randomization/random subset sampling	Standard RM-style uncertainty test
Predictor-label permutation	Tests whether specific physical channels matter
Seasonal/window-function surrogate	Tests whether observing cadence creates false peaks
Multiple-testing correction	Needed because many lags and targets are scanned

Without these, a referee may say that the long-lag synergy peaks are artifacts of red noise or windowing.

1.7. Compare SURD with standard reverberation methods

This is essential. The paper should not present SURD alone. It should compare:

ICCF lag, DCF/ZDCF lag, JAVELIN/DRW lag, SURD peak lag.

The goal is not necessarily to make them identical. The paper can say:

ICCF measures the dominant pairwise linear reverberation delay, whereas SURD measures how much information about the future target is uniquely, redundantly, or synergistically supplied by a set of past variables.

That distinction is very important. Reverberation mapping is a standard time-domain technique for estimating BLR size and structure from light echoes. SURD should be presented as a complementary diagnostic, not a replacement for classical RM.

1.8. A stronger synthetic-data section is needed

The synthetic tests should become a major validation section. I suggest four synthetic systems.

Synthetic model A: single-driver echo

$$C(t) = U(t) + \varepsilon_C(t),$$

$$L(t) = a U(t-\tau) + \varepsilon_L(t).$$

Expected result: high unique information from C, low synergy.

Synthetic model B: redundant proxies

$$C1(t) = U(t) + \varepsilon_1(t),$$

$$C2(t) = U(t) + \varepsilon_2(t),$$

$$L(t) = a U(t-\tau) + \varepsilon_L(t).$$

Expected result: high redundancy.

Synthetic model C: true nonlinear synergy

$$L(t) = a C1(t-\tau) C2(t-\tau) + \varepsilon_L(t).$$

Expected result: strong synergy.

Synthetic model D: two-zone BLR response

$$L(t) = a_1 C(t-\tau_1) + a_2 C(t-\tau_2) + \varepsilon_L(t).$$

Expected result: possibly multiple SURD features or broad lag structure.

Then repeat each model using:

- irregular sampling matching NGC 5548,
- real measurement errors,
- red-noise continuum,
- seasonal gaps,
- different bin numbers,
- different cadences.

This will make the method much more defensible.

1.9. The physical paper should focus on velocity-resolved H β

The strongest paper direction is not just “SURD on AGN light curves.” It is more specific:

Can SURD detect nonlinear and multi-component information flow among the continuum, blue wing, core, and red wing of H β in NGC 5548?

This is physically meaningful because velocity-resolved H β probes BLR kinematics. Recent and historical NGC 5548 studies discuss velocity-resolved lags and complex BLR behavior.

A good target-predictor table would be:

Experiment	Predictors	Target	Physical question
1	Continuum, blue wing	H β core	Does the core require both driving continuum and wing information?
2	Continuum, core	blue wing	Is the blue wing conditionally driven by the core?
3	Continuum, core	red wing	Does red-wing response differ from blue-wing response?
4	Continuum, red wing	blue wing	Search for asymmetry/inflow/outflow signatures

Experiment	Predictors	Target	Physical question
5	Blue wing, red wing	core	Does the line core combine information from both wings?

This could produce a genuinely novel result.

1.10. What the paper's main claim should be

Do not claim yet:

"We discovered the true H β lag is 111 days."

That would be too strong.

A safer and better claim is:

"We introduce SURD as a complementary information-theoretic diagnostic for AGN reverberation mapping. Applied to NGC 5548 H β velocity-resolved light curves, SURD reveals statistically significant velocity-dependent synergistic information flow that is not captured by pairwise correlation alone. The synergy peaks occur on longer timescales than standard pairwise reverberation lags, suggesting sensitivity to multi-component BLR dynamics, long-memory continuum variability, or nonlinear coupling among line components."

This is publishable if the statistics are strengthened.

1.11. Minimum extra work needed before writing the paper

I would do these tasks before preparing the manuscript:

1. Clean the notebook into a reproducible script/package. Remove hard-coded Colab paths, duplicated cells, old failed cells, and presentation-only cells.
2. Verify all time axes and MJD conversions. Especially the `.spc` filename conversion.
3. Use only physically meaningful predictor sets. Remove the duplicated-continuum result from the main paper, or present it only as a sanity check.
4. Add standard reverberation lag comparisons. ICCF, DCF/ZDCF, and ideally JAVELIN or another DRW-based method.
5. Add stronger null models. Especially block bootstrap, red-noise surrogates, and seasonal-window surrogates.
6. Add uncertainty bands. Report confidence intervals for unique, redundant, synergistic terms and information leak.
7. Test synthetic systems with known truth. Include single-driver, redundant, synergistic, and two-zone BLR examples.
8. Clarify the interpretation of information leak. Leakage is not simply "unexplained physics"; it may include noise, missing variables, wrong lag, finite-sample bias, binning bias, and stochastic variability.
9. Correct for multiple lag scans. Since many lags and targets are tested, significance must account for multiple comparisons.
10. Build a physical interpretation section. Compare blue/core/red behavior with inflow, outflow, virialized BLR, disk-wind, and multi-zone response possibilities.

1.12. Suggested paper structure

A strong paper could be organized as follows:

Title

Synergistic Information Decomposition of Velocity-Resolved AGN Reverberation Mapping: A Case Study of NGC 5548 H β

Abstract

State the problem, introduce SURD, validate on synthetic reverberation systems, apply to NGC 5548, report velocity-dependent synergy and leakage, and emphasize that SURD complements classical lag

methods.

1. Introduction

AGN reverberation mapping, limitations of pairwise correlation, need for multivariate information methods.

2. Data

NGC 5548 continuum, H β spectra, velocity-bin definitions, cadence, time range, uncertainties.

3. Method

SURD decomposition, lag construction, discretization/estimators, information leak, surrogate tests, uncertainty estimation.

4. Synthetic Validation

Single-driver, redundant-proxy, true-synergy, and two-zone BLR simulations.

5. Real Data Results

Continuum-H β results, velocity-resolved results, blue/core/red target analyses, robustness to cadence and binning.

6. Comparison with Classical Reverberation Mapping

ICCF/ZDCF/JAVELIN versus SURD.

7. Physical Interpretation

Long-lag synergy, velocity dependence, possible BLR multi-component structure, caution about red noise and hidden drivers.

8. Limitations

Finite sample, interpolation, discretization, stationarity, missing UV/X-ray driver, seasonal gaps.

9. Conclusions

SURD is a useful complementary diagnostic, but not a replacement for standard RM.

1.13. My final recommendation

The best next step is to turn the current work into a method-validation plus NGC 5548 case-study paper, not yet a strong discovery paper. The novelty should be methodological:

“SURD provides a new multivariate information-decomposition view of AGN reverberation mapping.”

The strongest publishable result will come if you show that, after rigorous synthetic tests and null models, the H β velocity components of NGC 5548 show statistically significant, velocity-dependent synergistic information flow beyond what ordinary pairwise lag analysis can reveal.

2. Other interesting AGN databases for this research

Yes. For a SURD / causal-information AGN paper, I would not use only one database. I would design the paper around several classes of AGN datasets, because each one tests a different physical problem.

2.1. Best AGN databases for your research direction

2.1.1. AGN Watch / classic reverberation-mapping datasets

This is the most natural continuation of your current NGC 5548 work. AGN Watch contains long-term continuum and emission-line monitoring for famous nearby Seyfert galaxies. For example, its historical NGC 5548 material includes UV continuum bands, optical continuum, and major emission lines such as Ly α , C IV, C III], N V, Si IV, He II, and H β .

Good targets:

Object	Why interesting for SURD
NGC 5548	benchmark object; many campaigns; velocity-resolved H β possible
NGC 4151	strong variability, complex absorption, nearby Seyfert
NGC 3783	good multi-line reverberation target
Fairall 9	clean broad-line AGN, useful comparison
Mrk 509	multiwavelength campaigns, X-ray/UV/optical coupling

Research idea:

Use SURD to compare whether different emission lines receive unique, redundant, or synergistic information from UV/optical continua.

Example target-predictor setup:

$$T = F_{C\ IV}(t+\tau),$$

$$S1 = F_{1350}(t), S2 = F_{5100}(t), S3 = F_{Ly\alpha}(t).$$

This asks whether C IV variability is driven uniquely by the UV continuum, redundantly by several continua, or synergistically by continuum plus another line.

2.1.2. AGN STORM / AGN STORM 2

This is probably the best high-quality dataset for a serious paper. AGN STORM provides dense, multiwavelength monitoring of NGC 5548, including UV, optical, X-ray, and emission-line light curves. A key point is that the far-UV continuum near 1350 Å is often treated as a better proxy for the ionizing continuum than optical 5100 Å, while optical light curves can be delayed and smoothed relative to the UV.

Why it is excellent for SURD:

Feature	Why useful
UV + optical continuum	test information flow across accretion-disk radii
X-ray + UV	test reprocessing versus intrinsic disk variability
many emission lines	compare BLR stratification
anomalous line response episodes	test nonstationary causal structure
velocity-resolved lines	ideal for blue/core/red SURD analysis

Very good research question:

Does the optical continuum provide unique information about H β once the UV continuum is known, or is it mostly redundant?

Mathematically:

$$T = H\beta(t+\tau),$$

$$S1 = F_{1350}(t), S2 = F_{5100}(t).$$

If S1 dominates uniquely, the UV is the main driver.

If S1 and S2 are redundant, both trace the same underlying continuum.

If synergy appears, H β may respond to a combination of inner and outer disk variability.

This is stronger than using only optical continuum.

2.1.3. SDSS Reverberation Mapping project

This is one of the best large-sample datasets. The SDSS-RM project contains multi-year photometric and spectroscopic light curves for hundreds of broad-line quasars. One recent SDSS-RM description reports 11 years of photometric and 7 years of spectroscopic light curves for 849 broad-line quasars over $0.1 < z < 4.5$.

The SDSS-V Black Hole Mapper RM program is also explicitly designed to measure continuum-emission-line time lags and black-hole masses for quasars over a wide range of luminosities

and redshifts.

Why it is good:

Advantage	Importance
Many AGN	statistical paper, not just one-object case study
multiple emission lines	H β , Mg II, C IV depending on redshift
spectra + photometry	allows continuum-line SURD
broad luminosity/redshift range	test physical trends

Possible paper:

“Information-decomposition reverberation mapping of SDSS quasars: how unique and synergistic information changes with luminosity, redshift, and emission line.”

Possible target variables:

$$T = F_{\text{Mg II}}(t+\tau),$$

or

$$T = F_{\text{C IV}}(t+\tau).$$

Predictors:

$$S1 = g(t), S2 = r(t), S3 = i(t),$$

or continuum windows extracted from spectra.

This would allow population-level results:

$$S_{\text{synergy}} = S_{\text{synergy}}(L_{\text{bol}}, z, M_{\text{BH}}, \lambda_{\text{Edd}}).$$

2.1.4. OzDES Reverberation Mapping

OzDES is very interesting because it is a high-redshift RM survey. The final OzDES RM work reports 6-7 years of photometric and spectroscopic observations of 735 AGN over $0.13 < z < 3.85$, with H β , Mg II and C IV reverberation information.

Why it is interesting:

Feature	Research value
high-redshift AGN	different luminosity regime from nearby Seyferts
H β , Mg II, C IV	compare line physics
long baseline	good for long lags
many objects	statistical SURD possible

Good research question:

Is the information decomposition different for H β , Mg II, and C IV?

For example:

$$T = F_{\text{line}}(t+\tau),$$

$$S1 = g(t), S2 = r(t), S3 = i(t).$$

Then compare:

$U_{\text{H}\beta}$, $R_{\text{H}\beta}$, $S_{\text{H}\beta}$,

$U_{\text{Mg II}}$, $R_{\text{Mg II}}$, $S_{\text{Mg II}}$,

$U_{\text{C IV}}$, $R_{\text{C IV}}$, $S_{\text{C IV}}$.

This could become a strong comparative paper on BLR stratification.

2.1.5. ZTF AGN light curves

ZTF is excellent for large-scale optical variability. It is not primarily a spectroscopic reverberation database, but it is very useful for AGN variability, changing-look AGN, flares, anomalous variability, and classification. Recent work continues to use ZTF for high-cadence AGN light curves and changing-look AGN studies.

Why useful for SURD:

Feature	Use
huge number of AGN	machine-learning/statistical paper
high cadence	variability causality and anomaly detection
g/r/i bands	multiband optical information flow
changing-look AGN	state-transition dynamics

Possible target-predictor setup:

$$T = r(t+\tau),$$

$$S1 = g(t), S2 = i(t).$$

Or for changing-look AGN:

$$T = \text{state transition indicator},$$

$$S1 = \text{g-band variability features}, S2 = \text{r-band variability features}, S3 = \text{WISE IR}.$$

Good paper idea:

“Can synergistic optical variability predict changing-look AGN transitions?”

This would be less reverberation mapping and more AGN variability/transition prediction.

2.1.6. ASAS-SN AGN light curves

ASAS-SN surveys the visible sky every night down to about 18th magnitude, and its Sky Patrol database provides public photometry/light curves. There are also ASAS-SN AGN light-curve studies, including SDSS AGN samples.

Why useful:

Feature	Use
all-sky coverage	long-term monitoring
bright nearby AGN	low-redshift Seyferts
easy light-curve access	good for pilot projects
long baselines	secular variability, changing-look events

Main limitation: usually only photometric bands, not emission-line light curves. So it is weaker for BLR reverberation, but good for continuum variability.

Good research idea:

Compare SURD decomposition of optical bands for normal AGN versus changing-look AGN.

2.1.7. Fermi-LAT 4LAC / blazar light curves

For blazars, Fermi-LAT is extremely interesting. The 4LAC-DR3 catalog is based on 12 years of gamma-ray data and includes spectral parameters, SEDs, yearly light curves and associations for 3407 high-Galactic-latitude AGN.

Why useful:

Feature	Use
gamma-ray AGN	jet-dominated physics
long-term light curves	flare dynamics
can combine with optical/radio	multiwavelength causality
blazar subclasses	FSRQ vs BL Lac comparison

Possible SURD setup:

$$T = F_{\gamma}(t+\tau),$$
$$S1 = F_{\text{optical}}(t), S2 = F_{\text{radio}}(t), S3 = F_{\text{X-ray}}(t).$$

Research question:

Are gamma-ray flares uniquely predicted by optical activity, or do they require synergistic information from optical plus radio/X-ray variability?

This would be a different, very attractive paper: SURD for jet variability, not BLR reverberation.

2.1.8. MACHO quasar light curves

The MACHO project has older but very long optical light curves for quasars behind the Magellanic Clouds. A public MACHO quasar sample contains light curves, spectra and finding charts for quasars behind the LMC and SMC.

Why useful:

Feature	Use
very long baseline	long-memory variability
two optical bands	simple multiband SURD
older independent dataset	good external validation
small sample	manageable pilot study

This is not the best main dataset, but it is useful as a robustness comparison.

2.2. My recommended priority order

For your specific research direction, I would choose this order:

Priority	Dataset	Best use
1	AGN STORM / STORM 2	high-quality multiband + multi-line physics
2	SDSS-RM	population-level SURD reverberation mapping
3	OzDES RM	high-redshift H β /Mg II/C IV comparison
4	AGN Watch	nearby Seyfert benchmark objects
5	Fermi-LAT 4LAC + optical/radio	blazar jet causal-information analysis
6	ZTF AGN	changing-look AGN and anomaly detection
7	ASAS-SN	bright nearby AGN long-term continuum variability
8	MACHO quasars	long-baseline validation sample

2.3. Best research-paper strategy

I suggest two papers, not one.

Paper 1: Reverberation-mapping SURD paper

Use:

AGN STORM + AGN Watch + SDSS-RM/OzDES.

Main question:

Can SURD reveal unique, redundant, and synergistic information flow between continuum bands and broad emission lines?

Best targets:

$$T = \text{H}\beta, \text{Mg II}, \text{C IV}.$$

Best predictors:

S1 = UV continuum, S2 = optical continuum, S3 = line component.

Paper 2: AGN state-change / jet-variability SURD paper

Use:

ZTF + ASAS-SN + Fermi-LAT.

Main question:

Can synergistic information predict AGN flares, changing-look transitions, or blazar gamma-ray activity?

This paper would be closer to machine learning / time-series causal discovery.

2.4. The most publishable immediate idea

The strongest next paper, in my opinion, is:

“Synergistic information decomposition of multiwavelength reverberation mapping in NGC 5548 and SDSS-RM quasars.”

Start with NGC 5548 because it is physically rich and well studied. Then test whether the same SURD patterns appear in a statistical sample from SDSS-RM or OzDES.

That combination gives you both:

physical interpretation

and

population-level statistical significance.

3. Source notes included from the original answers

The original answers referred to the following web sources and source families:

- Nature Communications article on SURD: "Decomposing causality into its synergistic, unique, and redundant components".
- NGC 5548 reverberation-mapping and velocity-resolved H β studies, including recent 2023 monitoring.
- General reverberation-mapping reviews and resources.
- AGN Watch historical data pages.
- AGN STORM / NGC 5548 multiwavelength monitoring papers.
- SDSS Reverberation Mapping and SDSS-V Black Hole Mapper RM program descriptions.
- OzDES Reverberation Mapping survey descriptions.
- ZTF AGN variability and changing-look AGN studies.
- ASAS-SN Sky Patrol and AGN light-curve resources.
- Fermi-LAT 4LAC-DR3 catalog resources.
- MACHO quasar light-curve resources.